

**School of Engineering and Applied Science (SEAS),
Ahmedabad University**

ECE501: Digital Image Processing

Group Name: Humans.exe

Project 2: Person Retrieval

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I. Introduction

- **Goal:** Finalize the classical person retrieval pipeline by integrating Fisherfaces (PCA + LDA), improving database organization, and enabling an interactive retrieval interface for testing.
- **Current Stage:** The system now includes complete data preprocessing, dimensionality reduction, discriminative feature extraction, and distance-based matching. The ORL dataset has been structured, validated, and successfully loaded for model training.
- **Focus of This Week:** Implementing Fisherfaces for high-accuracy classification, resolving dataset path and image-size inconsistencies, constructing PCA and LDA models, and building an interactive query-based retrieval interface using widgets for visualization.
- **Outcome:** A fully functional classical face retrieval system is achieved. The model can process query images, project them into PCA-LDA space, compute similarity scores, and display the top matching faces with improved accuracy and interpretability.

II. Objectives

The primary objective for this stage was to complete the Fisherfaces-based retrieval system and ensure a fully functional end-to-end classical person retrieval workflow. Specifically, the goals were:

- Implement the full PCA–LDA (Fisherfaces) pipeline to obtain discriminative feature projections for more accurate face matching.
- Resolve dataset-related issues, including directory inconsistencies, image resizing, and uniform vector formation for model training.
- Generate and store PCA and LDA feature representations for the entire ORL dataset to support efficient retrieval.
- Build an interactive interface using widgets to allow selection of query images and display of top retrieved matches.
- Ensure reliable similarity computation by projecting query images into the trained PCA–LDA subspace and comparing distances effectively.

III. What has been done so far (Progress)

The team successfully completed the key tasks planned for this week and finalized the classical Fisherfaces-based person retrieval system. The following milestones were achieved:

- **Fisherfaces Implementation:** Successfully integrated the PCA–LDA (Fisherfaces) pipeline, enabling discriminative feature extraction and significantly improving class separability.
- **Dataset Structure Cleanup:** Fixed directory inconsistencies in the ORL dataset, ensured correct detection of subject folders, and standardized all images to the required 112×92 resolution for stable PCA transformation.
- **Dimensionality Consistency:** Resolved feature mismatch errors between training and query images by enforcing uniform vector sizes and consistent preprocessing steps.
- **Retrieval Module Construction:** Computed and stored PCA and LDA projections for all images in the dataset, making similarity computation faster and more reliable.
- **Interactive Interface:** Developed an interactive widget-based interface allowing users to select a query image, perform real-time feature projection, and view the top retrieved matches with distances.
- **Improved Matching Accuracy:** Performed iterative testing and parameter adjustments, achieving more stable retrieval performance and reduced mismatches across multiple subjects.

- **Scenario Testing:** Conducted tests on various query images to verify system robustness, including cross-subject matching, lighting variations, and different facial expressions.

IV. What is planned for next week

- Presenting the model in front of the class.

References

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- [3] Lu, X. (2003). Image analysis for face recognition. *Personal Notes*, 36.
- [4] Shah, J. H., Sharif, M., Raza, M., & Azeem, A. (2013). A Survey: Linear and Non-linear PCA Based Face Recognition Techniques. *International Arab Journal of Information Technology*, 10(6), 536–545.